# The Second Corpus Study on German Nominal Compounding Constraints

## Background

German nominal compounding is a notoriously unpredictable and irregular process. About 40% of German N+N compounds insert a so-called linking element that is not confidently predictable for most of constituent classes. For example, nouns Birne ‘pear’ and Kirsche ‘cherry’ build compounds Birne-n-saft ‘pear juice’ but Kirsch-0-saft ‘cherry juice’ despite having nearly identical phonological, morphological, and semantic properties. The only linguistic generalizations available to gauge the correct linker are (mostly) irregular constraints that are eagerly violated by numerous compounds.

Our first corpus study on German nominal compounding collected 64 constraints from the linguistic literature and verified 39 of them against ~280k German N+N compounds. The corpus study demonstrated that most of the constraints were irregular, indeed. The most crucial funding was that not all of the constraints were confirmed on the massive data. Specifically, eight constraints were only partially supported by the data, and three were not supported at all.

The main limitation of the first corpus study was that none of the constraints involving semantic or lexical conditions could be implemented. The present study fills in most of the gaps remaining after the first corpus study:

1. It implements 17 constraints involving semantics. Fourteen of the 17 constraints are pluralic constraints actively discussed in recent linguistics literature.
2. It applies more detailed statistic analyses to analyze relations between different ontological properties of the constraints (application manner, types of linguistic conditions etc.) and its coverage and regularity measures.
3. Additionally, it implements a couple of further constraints and verifies the constraints on further datasets.

## Datasets

### *GeCoDB_v06*

> 282767 compound lemmata. 3614 N1 lemmata. 9 linkers, 11 linker allomorphs. 4768 unique N1+linker combinations.

~280k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at least 20 in DeReKo. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Database/compounds/decow16ax_comps_counts.txt'), with further processing steps in Shmalts BA Thesis, final preprocessing in Shmalts 2026.

The dataset is used to (re-)verify 40 non-semantic constraints.

### *GeCoDB_v07*

> 275411 compound lemmata. 3593 N1 lemmata. 10 linkers, 12 linker allomorphs. 4741 unique N1+linker combinations.

~275k German N+N compounds from *GeCoDB_v06* whose constituents are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06*.

### *GeCoDB_v07_GN*

> 48976 compound lemmata (48660 unique N1+N2 combinations). 3319 N1 lemmata. 10 linkers, 12 linker allomorphs. 3585 unique N1+linker combinations.

~49k German N+N compounds from *GeCoDB_v07* that are present in GermaNet.

The dataset is used to verify the same 40 non-semantic constraints as *GeCoDB_v06* and five further semantic non-pluralic constraints.

### *SP18_Pl_Constr*

> 2294 compound lemmata. 43 N1 lemmata. 8 linkers, 9 linker allomorphs. 76 unique N1+(Sg/Pl meaning) combinations. 86 unique N1+linker combinations.

~2.3k German N+N compounds from the DECOW16A corpus that are built from nouns found in CELEX2 and have a WC of at

least 20 in DeReKo, with annotated singular/plural meaning of N1. The dataset originates from Schäfer & Pankratz 2018 (file 'linkingelements/Data/Corpusstudy/Concordance.full.csv').

The dataset is used to verify 12 pluralic constraints.

## Outline

1. Chapter 1. We first inspect the ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language: application manner, type(s) of accessed linguistic properties, application item. The results corroborate and slightly elaborate the finding from the first corpus study.
2. Chapter 2. We then inspect the impact of data and metric choice on surface constraint applicability. We find that the constraint applicability values are statistically very similar both across the *GeCoDB* dataset versions and the item/type variants of the metrics. Based on these results, we focus on the results of *GeCoDB_v07_GN* and *SP18_Pl_Constr* and only concentrate on the item variant of the applicability statistics.
3. Chapter 3. We analyze the values of application measures of the constraints on the flattened applicability statistics, which includes their distribution and correlation. We establish relation between the values of coverage and regularity of the constraints, and their ontological properties from step 1.
4. Chapter 4. We finally introduce and apply the new (irregularity) scale to our results.

## Results

### 1. Constraint Ontological Properties

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. Al but two out of 65 constraints do not access the linguistic properties of N1.

### 2. Constraint Applicability Vs. Datasets & Metrics

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated between *GeCoDB_v06* and *GeCoDB_v07* (Wilcoxon signed-rank test all p-values > $0.1$). *GeCoDB_v07_GN* appears to produce greater item regularity values, but only on the L2P constraints (Wilcoxon signed-rank test p-value $0.004$ as compared with *GeCoDB_v06*). Since all not supported constraints from the original study were P2L constraints -- where *GeCoDB_v06* and *GeCoDB_v07_GN* produce statistically identical values (Wilcoxon signed-rank test p-value $0.268$). That speaks in favor of the validity of the original results and attests the overall robustness of the constraint applicability statistics. At the same time, it points out to a bigger permissiveness of the original dataset when it comes to linker restrictions. We choose to proceed with *GeCoDB_v07_GN* as the middle ground between trustworthiness and consistency with the original corpus study.
- The item and type variants of both applicability metrics are statistically very similar (Wilcoxon signed-rank test p-values $0.085$, $0.993$, respectively), which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

### 3. Constraint Applicability

- Most of the item coverage values are very low at $<0.05$. The item regularity values are more even. They are clustered naturally by the thresholds of $0.7$ and $0.9$. We propose a new (ir)regularity scale of $[0.5; 0.7)$, $[0.7; 0.9)$, $[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints (Mann-Whitney U-test p-value $0.017$), while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- In lines to the claim in Shmalts 2026, correlation between item coverage and regularity is negligible (Spearman rho $-0.135$).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.2$ at intercept $0.83$, p-value $0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

[4. Constraint Support, Productivity & Regularity]

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 downgraded to partially support at the new scale. The new *GeCoDB* versions produce an even better result on the old constraints due to cleaner data. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*. On top of that, the changed checkers for the constraints triggered by morphological and phonological complexity of the N1 led to these two constraints being partially supported and unsupported, respectively.
- Ten constraints out of 48 turn out to be unproductive by the Tolerance Principle. Regular constraints by the ITP constitute half of the constraint set: 24 out of 48.
- We find that no constraints require a change of definition. Only regularity levels should be adjusted for them.

# 1. Constraint Ontological Properties

## Description

In this chapter, we are interested in ontological properties of constraints as linguistic generalizations, as opposed to their empirical application in the language. More specifically, we will inspect the following properties of the constraint set:

1. The distribution of the constraints by the application manner.
2. The distribution of the constraints by the type(s) of accessed linguistic properties (domain type). Note that the default constraint is excluded from consideration in this step.
3. The distribution of the constraints by the application item.

**NB!** In this chapter, we first take a look at the entire set of constraints, whether they are confirmed or not.

## Results

The results corroborate and slightly elaborate the finding from the first corpus study.

- Morphology constitutes the core of German nominal compounding, followed by derivation and phonology. Semantics, while contributing to a considerable number of constrains, plays a complementary role and is almost negligible when isolated. Lexical and most of mixed constraints are underrepresented.
- N1 stays the cornerstone of constraint definitions. All but two out of 65 constraints do not access the linguistic properties of N1.

1. In total, 65 constraints are summarized in this corpus study. The P2L constraints are 1.5 more than the L2P constraints.

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| | N |
|---|---|
| **total** | 65 |
| **P2L** | 39 |
| **L2P** | 25 |

2. We can clearly see that the morphological (including mixed) constraints constitute the core of the constraint set, making almost a quarter of all constraints. If we sum up all constraint involving morphology (so mixed as well), this number rises to roughly a half. Derivational and phonological constraints occupy a substantial share of the set as well. Semantics is negligible when isolated but makes $14.1\%$$14.1\%$ of all constraints when coupled with morphology. Lexical and further mixed constraints are overall rare. The same distribution holds as well for the P2L and the L2P subsets of constraints, except for lexical constraints being the second most frequent type of L2P constraints, while the number of lexical P2L constraints is one.

The drop in domain type frequency is overall quite gradual, with no large gaps: $18.8\%$$18.8\%$, $17.2\%$$17.2\%$, $14.1\%$$14.1\%$, $9.4\%$$9.4\%$, then $7.8\%$$7.8\%$, etc. Notably, the constraints are distributed much more evenly by the linguistic domain type among the P2L constraints than among the L2P constraints. The P2L subset resembles the entire constraint set in this regard, with a gradual decrease in domain type frequency. On the contrary, domain type frequency quickly drops to low values in the L2P subset, whose top two domain types (morphological and lexical) $41.7\%$$41.7\%$ of all L2P constraints. These are then followed by a long tail of underrepresented types.

The L2P constraints almost ignore semantics. That means that German nominal compounding does not have formal semantic restrictions that would prohibit some linkers from appearing in certain semantic contexts (as opposed to, say, morphological formal restrictions). The P2L constraints, on the contrary, draw a lot on semantics. This, and the fact that most of the semantics-involving constraints are coupled with a morphological condition, indicates that semantics only has an complementary function in German nominal compounding. It can only serve as a soft clue when generating compounds but cannot really restrict linker choice.

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| | total | | p2l | | l2p | |
|---|---|---|---|---|---|---|
| | N | % | N | % | N | % |
| **morphological** | 12 | 18.8 | 6 | 15.8 | 6 | 25.0 |
| **derivational** | 11 | 17.2 | 8 | 21.1 | 3 | 12.5 |
| **morphological+semantic** | 9 | 14.1 | 8 | 21.1 | 1 | 4.2 |
| **phonological** | 6 | 9.4 | 3 | 7.9 | 3 | 12.5 |
| **morphological+phonological+semantic** | 5 | 7.8 | 4 | 10.5 | 1 | 4.2 |
| **lexical** | 5 | 7.8 | 1 | 2.6 | 4 | 16.7 |
| **morphological+phonological** | 3 | 4.7 | 2 | 5.3 | 1 | 4.2 |
| **derivational+phonological** | 3 | 4.7 | 0 | 0.0 | 3 | 12.5 |
| **derivational+morphological** | 2 | 3.1 | 1 | 2.6 | 1 | 4.2 |
| **derivational+semantic** | 2 | 3.1 | 2 | 5.3 | 0 | 0.0 |
| **semantic** | 2 | 3.1 | 2 | 5.3 | 0 | 0.0 |
| **derivational+morphological+phonological** | 1 | 1.6 | 1 | 2.6 | 0 | 0.0 |
| **lexical+semantic** | 1 | 1.6 | 1 | 2.6 | 0 | 0.0 |
| **lexical+morphological+phonological+semantic** | 1 | 1.6 | 0 | 0.0 | 1 | 4.2 |
| **lexical+morphological** | 1 | 1.6 | 0 | 0.0 | 1 | 4.2 |

The conclusions made above are supported by the individual contributions of linguistic domains. By the individual contribution of a domain, we mean the total number of constraints that access linguistic properties from this domain, both mixed or not. For example, the individual contributions in a small set of three constraints [morphological, mophological+phonological, derivational+phonological] are: morphology: 2, phonology: 2, derivation: 1.

We can clearly see the overwhelming contribution of morphology. Slightly more that half of all constraints have a morphological condition. Semantics, derivation, and morphology have almost equal contributions, being present in almost a third of constraints each.

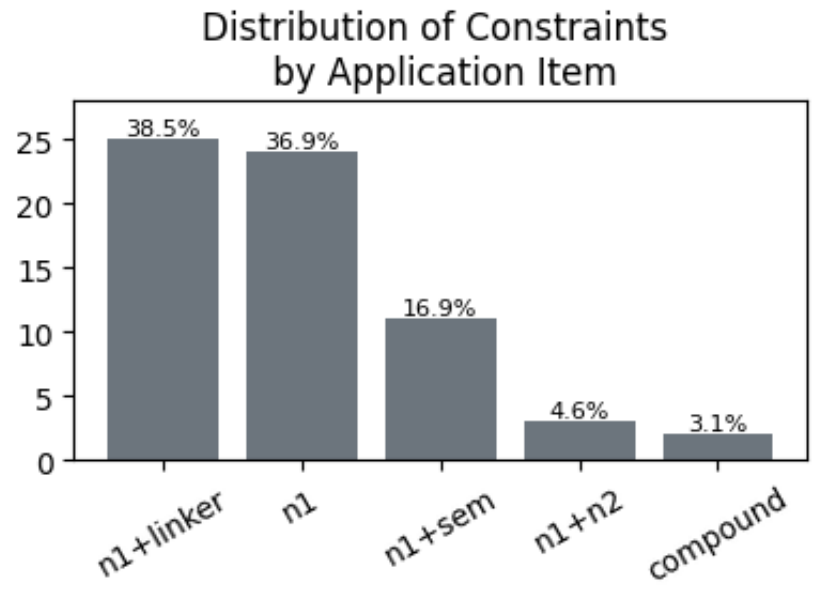
While the counts for unique domain types among the P2L and the L2P differs significantly, the individual contribution of each considered linguistic domain in not that different. Thus, morphology, phonology, and derivation contribute comparably to P2L and L2P constraints. However, lexical constraints are prevalent among the L2P constraints, and semantic ones -- among the P2L constraints, as discussed above. For instance, semantic constraint contribute to $44.7\%$$44.7\%$ of the P2L but only to $12.5\%$$12.5\%$ of the L2P constraints. The absolute numbers are even worse: $17$$17$ vs $3$$3$.

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| | total | | p2l | | l2p | |
|---|---|---|---|---|---|---|
| | N | % | N | % | N | % |
| **morphological** | 34 | 53.1 | 22 | 57.9 | 12 | 50.0 |
| **semantic** | 20 | 31.2 | 17 | 44.7 | 3 | 12.5 |
| **derivational** | 19 | 29.7 | 12 | 31.6 | 7 | 29.2 |
| **phonological** | 19 | 29.7 | 10 | 26.3 | 9 | 37.5 |
| **lexical** | 8 | 12.5 | 2 | 5.3 | 6 | 25.0 |

3. N1 plays the most prominent role in constraint definitions. Being the most frequent type when isolated, it also constitutes part of the further most frequent items: N1 + linker, N1 + plural semantics, etc. Actually, only two out of 65 constraints do not access the properties of N1.

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Below are the counts of unique items for each item type in each dataset.

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| | compound | n1 | n1+sem | n1+n2 | n1+linker |
|---|---|---|---|---|---|
| **gecodb_v06** | 282767 | 3614 | 0 | 0 | 4870 |
| **gecodb_v07** | 275411 | 3593 | 0 | 0 | 4741 |
| **gecodb_v07_gn** | 48976 | 3319 | 0 | 48660 | 3585 |
| **sp18_pl_constr** | 2294 | 0 | 76 | 0 | 86 |

## 2. Constraint Applicability Vs. Datasets & Metrics

### Description

Before we proceed to the analysis of coverage and regularity of the constraints, we need to inspect the impact of data and metric choice on surface constraint applicability.

1. We want to understand how robust the findings of the first corpus study are, and whether the (unexpected) irregularity of some constraints also extends to further dataset and is not a consequence of data choice.
2. We evaluate the difference between the item and type variant of coverage and regularity.

To assess that, we will run some statistical tests over the results obtained across different versions of the *GeCoDB* dataset, and across the item/type variant of applicability metrics.

**NB!** In this chapter, we take a look at the entire set of constraints, whether they are confirmed or not.

### Results

- The values of both item and type coverage and regularity come from an identical distribution and are highly correlated between *GeCoDB_v06* and *GeCoDB_v07* (Wilcoxon signed-rank test all p-values > $0.1$$0.1$). *GeCoDB_v07_GN* appears to produce greater item regularity values, but only on the L2P constraints (Wilcoxon signed-rank test p-value $0.004$$0.004$ as compared with *GeCoDB_v06*). Since all not supported constraints from the original study were P2L constraints -- where *GeCoDB_v06* and *GeCoDB_v07_GN* produce statistically identical values (Wilcoxon signed-rank test p-value $0.268$$0.268$). That speaks in favor of the validity of the original results and attests the overall robustness of the constraint applicability statistics. At the same time, it points out to a bigger permissiveness of the original dataset when it comes to linker restrictions. We choose to proceed with *GeCoDB_v07_GN* as the middle ground between trustworthiness and consistency with the original corpus study.
- The item and type variants of both applicability metrics are statistically very similar (Wilcoxon signed-rank test p-values $0.085$$0.085$, $0.993$$0.993$, respectively), which supports the equivalent finding from the first corpus study. We choose to proceed with the item variant as the more balanced version of the metrics.

Out of 65 constraints, we have the applicability statistics for 54. The application manner of those is nearly identical to that of the entire constraint set: the P2L constraints is almost 1.5 times larger than the L2P subset. From the missing 11 constraints, three had no covered compounds in the datasets (one in *GeCoDB*'s and two in *SP18_Pl_Constr*), and the remaining eight are not implemented. These all include semantic or lexical conditions.

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| | total | gecodb_v06 | gecodb_v07 | gecodb_v07_gn | sp18_pl_constr |
|---|---|---|---|---|---|
| **total** | 54 | 39 | 39 | 44 | 11 |
| **p2l** | 33 | 20 | 20 | 24 | 9 |
| **l2p** | 20 | 18 | 18 | 19 | 1 |

Draw the plots of item and type coverage and regularity values across the datasets and metrics.

1. One can clealy see that many values from different versions of the *GeCoDB* dataset are located at the same locations. For each unique pair of these dataset versions, we run a Wilcoxon signed-rank test to test whether the values of item and type coverage and regularity across the pair are at the same location (we consider only shared constraints, since *GeCoDB_v07_GN* verifies a few more). We choose the Wilcoxon signed-rank test because we are dealing with paired samples that are distributed abnormally. We additionally calculate the Spearman correlation coefficient to test for the mutual predictability of the values. We choose Spearman because, again, our values are distributed abnormally.

The results reveal that item and type regularity and coverage values of *GeCoDB_v06* and *GeCoDB_v07* are statistically identical. None of the comparisons reach a significant result of the Wilcoxon signed-rank (all p-values $>0.1$$>0.1$; two-sided alternative hypothesis: the locations are shifted). Moreover, the Spearman correlation coefficient is in a range of $(0.98; 1.0]$$(0.98; 1.0]$ for all comparisons.

On the contrary, both these versions are at odds with *GeCoDB_v07_GN*: Wilcoxon signed-rank test p-values are significant at $0.026$$0.026$, $0.023$$0.023$, respectively (two-sided alternative hypothesis: the locations are shifted). An additional analysis tells us that the values of *GeCoDB_v07_GN* are statistically larger (Wilcoxon signed-rank test p-values $0.013$$0.013$, $0.011$$0.011$, respectively, greater alternative hypothesis: the location of the former is greater).

If we inspect the source of the divergence, we realize that the difference is only statistical between the L2P but not the P2L constraints. We have learned that *GeCoDB_v06* and *GeCoDB_v07* are statistically identical so we may concentrate on one of them for this additional comparison. Let us proceed with *GeCoDB_v06* as the more divergent dataset for a stricter analysis. Indeed, comparing *GeCoDB_v07_GN* to *GeCoDB_v06* at the P2L and the L2P constraints only results in Wilcoxon signed-rank test p-values of $0.268$$0.268$ but $0.004$$0.004$, respectively (greater alternative hypothesis: the location of the former is greater). All other alternative hypothesis result in insignificant p-values. Note, however, that the difference holds for item regularity only and does not extend to type regularity (or any coverage variant), both between all and L2P only constraints. Moreover, even given this difference, the Spearman correlation stays very strong in range $(0.94; 1.0]$$(0.94; 1.0]$ for every measure even between *GeCoDB_v07_GN* and the other two.

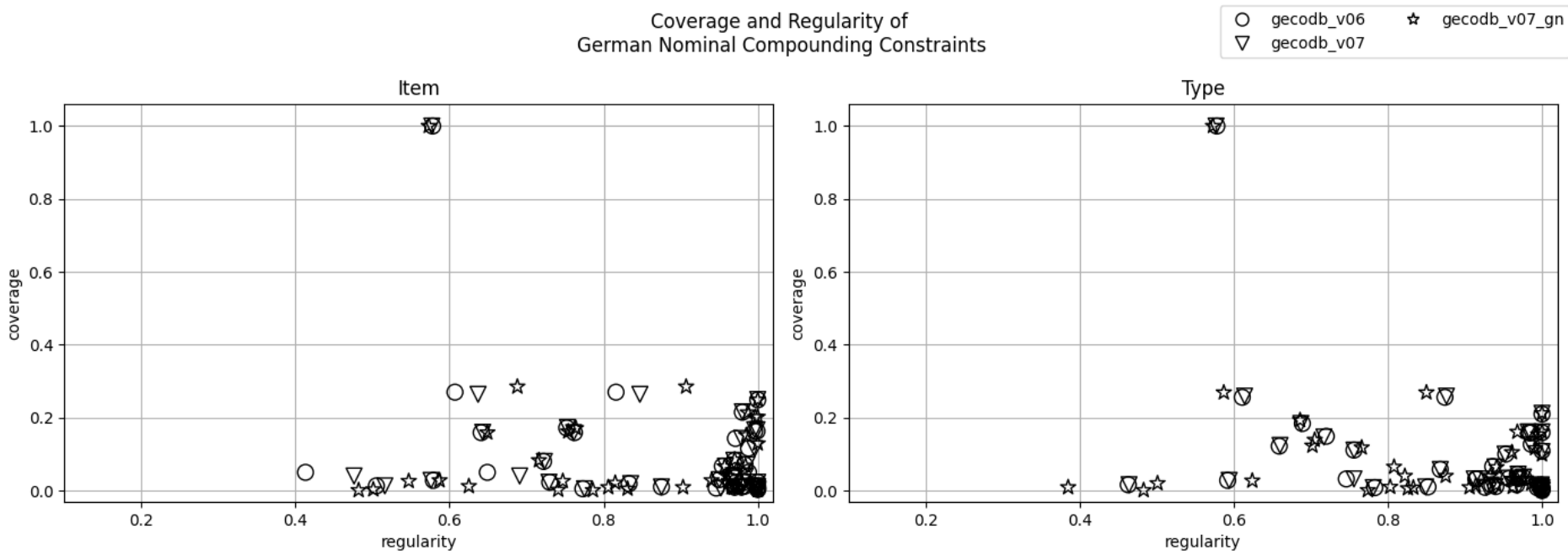
Please note that all the not supported constraints from Shmalts 2026 were P2L constraints. Given the close statistical similarity between the P2L constraints between *GeCoDB_v07_GN* and the originally used *GeCoDB_v06*, we conclude that these constraints would have been not supported in their original definitions over *GeCoDB_v07_GN*, either. While one might have argued that unexpectedly low regularity of some P2L constraints found in Shmalts 2026 was a result of the presence of low-quality data from the web in *GeCoDB_v06* (typos, unconventional compounds, etc.), compounds from GermaNet, that produce the same results, can hardly be accused of that.

Partially supported were both P2L and L2P constraints, so some partially supported L2P constraints would have been supported on *GeCoDB_v07_GN*.

On the whole, we interpret the findings the following way. The fact the results from *GeCoDB_v07_GN* are statistically identical to the results of *GeCoDB_v06* on the P2L constraints -- which are both the majority type, and the only not supported constraints -- provide additional evidence that the original *GeCoDB* versions was and is sufficiently clean and eligible for the corpus study, and that the results of the first corpus study are valid and legitimate. At the same time, statistically greater values of the L2P constraints over *GeCoDB_v07_GN* indicate that the original dataset was more permissive when it comes to linker restrictions*. We conclude that the datasets are interchangeable as long as one is mindful of the higher linker restriction strictness of *GeCoDB_v07_GN*. We choose to proceed with *GeCoDB_v07_GN* as the middle ground between trustworthiness and consistency with the original corpus study.

* However, since the result holds only for item and not type regularity, we suppose that *GeCoDB_v07_GN* is just more consistent in N1 compositionality but not crucially different in general.

```
g_dnames = [k for k in data_specs if k.startswith("gecodb")]
cvg_reg_plot(g_dnames)
```




```
# pairwise comparisons, alternative hypothesis: two-sided
```

| | | reg | | cvg | |
|---|---|---|---|---|---|
| | | item | type | item | type |
| comparison | | | | | |
| gecodb_v06 vs. gecodb_v07 | W | 227.500 | 274.000 | 185.000 | 128.500 |
| | p | 0.115 | 0.653 | 0.622 | 0.106 |
| | r | 0.988 | 0.999 | 0.999 | 1.000 |
| gecodb_v06 vs. gecodb_v07_gn | W | 220.000 | 297.500 | 297.000 | 321.000 |
| | p | 0.026 | 0.251 | 0.356 | 0.346 |
| | r | 0.948 | 0.985 | 0.968 | 0.987 |
| gecodb_v07 vs. gecodb_v07_gn | W | 216.500 | 295.000 | 302.000 | 366.000 |
| | p | 0.023 | 0.237 | 0.326 | 0.850 |
| | r | 0.955 | 0.987 | 0.970 | 0.987 |

```
# pairwise comparisons, alternative hypothesis: greater
```

| | | reg | | cvg | |
|---|---|---|---|---|---|
| | | item | type | item | type |
| comparison | | | | | |
| gecodb_v07_gn vs. gecodb_v06 | W | 539.000 | 461.500 | 297.000 | 456.000 |
| | p | 0.013 | 0.126 | 0.822 | 0.173 |
| gecodb_v07_gn vs. gecodb_v07 | W | 542.500 | 464.000 | 302.000 | 393.000 |
| | p | 0.011 | 0.119 | 0.837 | 0.425 |

```
# GeCoDB_v07_GN vs. GeCoDB_v06, alternative hypothesis: two-sided
```

| | | reg | | cvg | |
|---|---|---|---|---|---|
| | | item | type | item | type |
| comparison | | | | | |
| P2L | W | 118.500 | 145.500 | 72.000 | 115.000 |
| | p | 0.268 | 0.063 | 0.660 | 0.354 |
| L2P | W | 142.500 | 87.000 | 75.000 | 104.000 |
| | p | 0.004 | 0.346 | 0.677 | 0.204 |

2. The comparison of the item and type variants of the metrics allows to conclude that the two variants are statistically very similar. We choose to proceed with the item variant as the more balanced version of the metrics.

First, the central tendency measures between the two variants are almost identical for both coverage and regularity.

```
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| | | count | mean | std | min | 50% | 90% | max |
|---|---|---|---|---|---|---|---|---|
| reg | item | 54.0 | 0.812 | 0.223 | 0.088 | 0.924 | 1.000 | 1.0 |
| | type | 54.0 | 0.826 | 0.213 | 0.154 | 0.924 | 1.000 | 1.0 |
| cvg | item | 54.0 | 0.092 | 0.158 | 0.001 | 0.032 | 0.210 | 1.0 |
| | type | 54.0 | 0.086 | 0.159 | 0.000 | 0.028 | 0.206 | 1.0 |

Furthermore, the results of the Wilcoxon signed-rank test suggest that the two variants of both coverage and regularity come from the same distribution ($0.085$$0.085$ and $0.993$$0.993$, respectively; two-sided alternative hypothesis: the locations are not shifted). The very high Spearman correlation coefficients ($0.865$$0.865$ and $0.933$$0.933$ for coverage and regularity, respectively) and very low MAEs ($0.021$$0.021$ and $0.057$$0.057$ for coverage and regularity, respectively) confirm that the two variants of the metrics are interchangeable.

This finding quantitatively support the equivalent qualitative assumption from Shmalts 2026.

```
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```

| | cvg | reg |
|---|---|---|
| W | 535.000 | 696.000 |
| p | 0.085 | 0.993 |
| r | 0.865 | 0.933 |
| mae | 0.021 | 0.057 |

# 3. Constraint Applicability

## Description

This chapter is the core part of the notebook. It analyses the applicability values of the constraints, which includes their distribution and correlation, and seeks to establish a relation between the values of coverage and regularity of the constraints, and their ontological properties considered in chapter 1.

1. First, we qualitatively analyze the distribution of the applicability values of the constraints. We observe a natural clustering of the item coverage and regularity values, and propose a new (ir)regularity scale based on it.
2. We compare the item coverage and regularity values of the P2L vs. L2P constraints. The L2P constraints appear to be statistically more regular. The difference in coverage is unexpectedly insignificant.
3. We test for a correlation between coverage and regularity. We find that there exists a small negative correlation between both item and type coverage and regularity.
4. We investigate the effect of the domain type of constraints on their item regularity. We find that semantics is the only domain that has a significant effect on item regularity. As expected, the effect is negative.

**NB!** In this chapted, we filter out all unconfirmed constraints (item regularity of $<0.5$$<0.5$) and constraints with $<3$$<3$ covered items.

## Results

- Most of the item coverage values are very low at $<0.05$$<0.05$. The item regularity values are more even. They are clustered naturally by the thresholds of $0.7$$0.7$ and $0.9$$0.9$. We propose a new (ir)regularity scale of $[0.5; 0.7)$$[0.5; 0.7)$, $[0.7; 0.9)$$[0.7; 0.9)$, $[0.9; 1.0]$$[0.9; 1.0]$.
- We find that the L2P constraints appear to be statistically more regular than the P2L constraints (Mann-Whitney U-test p-value $0.017$$0.017$), while there is no observed statistical difference between the item coverage values across the two subsamples. The former corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirms its robustness after adding semantic constraints. The latter contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.
- In lines to the claim in Shmalts 2026, correlation between item coverage and regularity is negligible (Spearman rho $-0.135$$-0.135$).
- We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.2$$-0.2$ at intercept $0.83$$0.83$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

First, we only keep constraints that have at least $3$$3$ covered items and an item regularity of $\geq 0.5$$\geq 0.5$. That results in $49$$49$ constraints. Again, the number of the P2L constraints are almost 1.5 times the number of the L2P constraints.

```
Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.
```

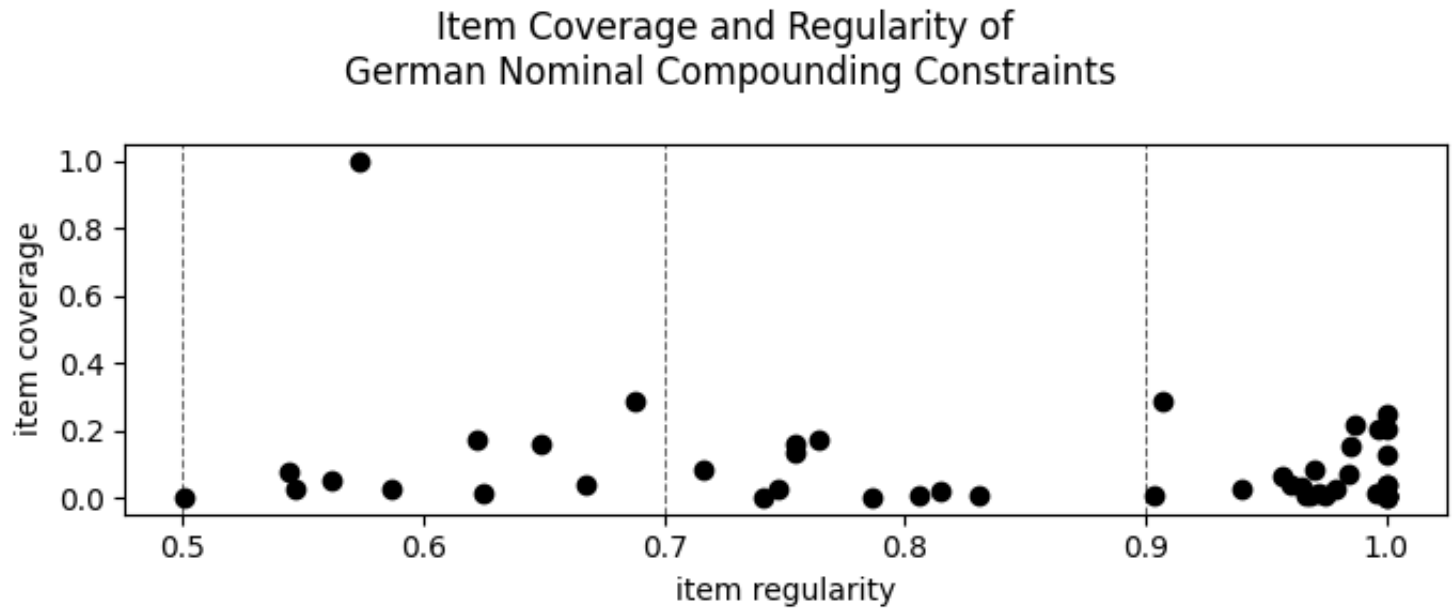
| | N |
|---|---|
| **total** | 49 |
| **p2l** | 29 |
| **l2p** | 19 |

1. As already reported in Shmalts 2026, the item coverage values are mostly very low, with more than half of the constraints lying in the item coverage range of $[\sim 0.0; 0.05)$$[\sim 0.0; 0.05)$ (29 of 49, or $59.2\%$$59.2\%$).

What concerns the item regularity values, the present study produces novel results. Specifically, it seems that the thresholds of $0.7$$0.7$ and $0.9$$0.9$ are the natural barriers between "low", "moderate", and "high" regularity. Indeed, the respective clusters are seen clearly on the scatter plot. The legitimacy of $0.9$$0.9$ is supported by the fact that it cuts through the largest gap between two neighboring item regularity values in the dataset. Among the values below $0.9$$0.9$, the threshold of $0.7$$0.7$ divides the values roughly on half (resulting in $11$$11$ and $10$$10$ values below and above, respectively).

The item regularity values clustering calls the original regularity scale from Shmalts 2026 into question. We propose the newly discovered thresholds of $0.7$$0.7$ and $0.9$$0.9$ as the new (ir)regularity scale. More on completion of the new scale with a "regular" level follows in chapter 4.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.




Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | N |
|---|---|
| **cvg: [0.0-0.05)** | 29 |
| **cvg: [0.05-0.2)** | 13 |
| **cvg: [0.2-1.01)** | 7 |
| **reg: [0.5-0.7)** | 11 |
| **reg: [0.7-0.9)** | 10 |
| **reg: [0.9-1.01)** | 28 |

The largest gap between two neighboring item regularity values is $0.831$$0.831$-$0.903$$0.903$. It is cuts precisely by the threshold of $0.9$$0.9$.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | left | right | diff |
|---|---|---|---|
| **0** | 0.831 | 0.903 | 0.072 |
| **1** | 0.501 | 0.544 | 0.043 |
| **2** | 0.587 | 0.622 | 0.035 |
| **3** | 0.907 | 0.940 | 0.033 |
| **4** | 0.688 | 0.716 | 0.028 |
| **5** | 0.716 | 0.741 | 0.025 |
| **6** | 0.625 | 0.649 | 0.024 |
| **7** | 0.764 | 0.786 | 0.022 |
| **8** | 0.667 | 0.688 | 0.021 |
| **9** | 0.786 | 0.806 | 0.020 |

2. The L2P constraints appear to be statistically more regular than the P2L constraints. This conclusion corroborates and quantitatively supports the equivalent finding from Shmalts 2026 and confirm its robustness after adding semantic constraints. On the contrary, there was no observed statistical difference between the item coverage values of the P2L vs. the L2P constraints. This finding contradicts the claim about item coverage difference made in the first corpus study and disproves it quantitatively.

Almost all L2P applicability values have a very low coverage and a high regularity, while the applicability values of the P2L constraints are distributed more or less evenly. Thus, 13 out of 19, or $68.4\%$$68.4\%$ of the L2P constraints have an item coverage $<0.05$$<0.05$. For the P2L constraints, this number is $55.2\%$$55.2\%$. The item regularity values are concentrated much closer to $1.0$$1.0$ among the L2P but not the P2L constraints. There are only two L2P constraints with item regularity $<0.7$$<0.7$. On the contrary, there are eight, or $27.6\%$$27.6\%$, such P2L constraints. Almost all L2P constraints are highly regular ($\geq 0.9$$\geq 0.9$): 14 out of 19, or $73.7\%$$73.7\%$. The P2L constraints do not even come close with mere $48.3\%$$48.3\%$ of highly regular constraints.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | all | P2L | L2P |
|---|---|---|---|
| **cvg: [0.0-0.05)** | 29 | 16 | 13 |
| **cvg: [0.05-0.2)** | 13 | 11 | 2 |
| **cvg: [0.2-1.01)** | 7 | 2 | 4 |
| **reg: [0.5-0.7)** | 11 | 8 | 2 |
| **reg: [0.7-0.9)** | 10 | 7 | 3 |
| **reg: [0.9-1.01)** | 28 | 14 | 14 |

We now run a Mann-Whitney U-test over the applicability values of the P2L vs. the L2P constraints. We choose the Mann-Whitney U-test because we are dealing with two independent samples whose distributions are abnormal and skewed. Since we suppose that the L2P constraints have lower coverage and higher regularity, we choose the two one-tailed alternative hypotheses: greater and less.

We can clearly see that the L2P constraints are undeniably more regular than the P2L constraints at sig = 0.05 (p-value $0.017$$0.017$; one-tailed alternative: L2P item regularity values are greater). The difference in coverage is not observed, with a p-value of $0.143$$0.143$ (one-tailed alternative: P2L item coverage values are greater).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| alternative | | cvg | reg |
|---|---|---|---|
| P2L>L2P | U | 326.500 | 175.000 |
| | p | 0.143 | 0.984 |
| P2L<L2P | U | 326.500 | 175.000 |
| | p | 0.862 | 0.017 |

3. In lines to the claim in Shmalts 2026, correlation between item coverage and regularity is negligible (Spearman rho $-0.135$$-0.135$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

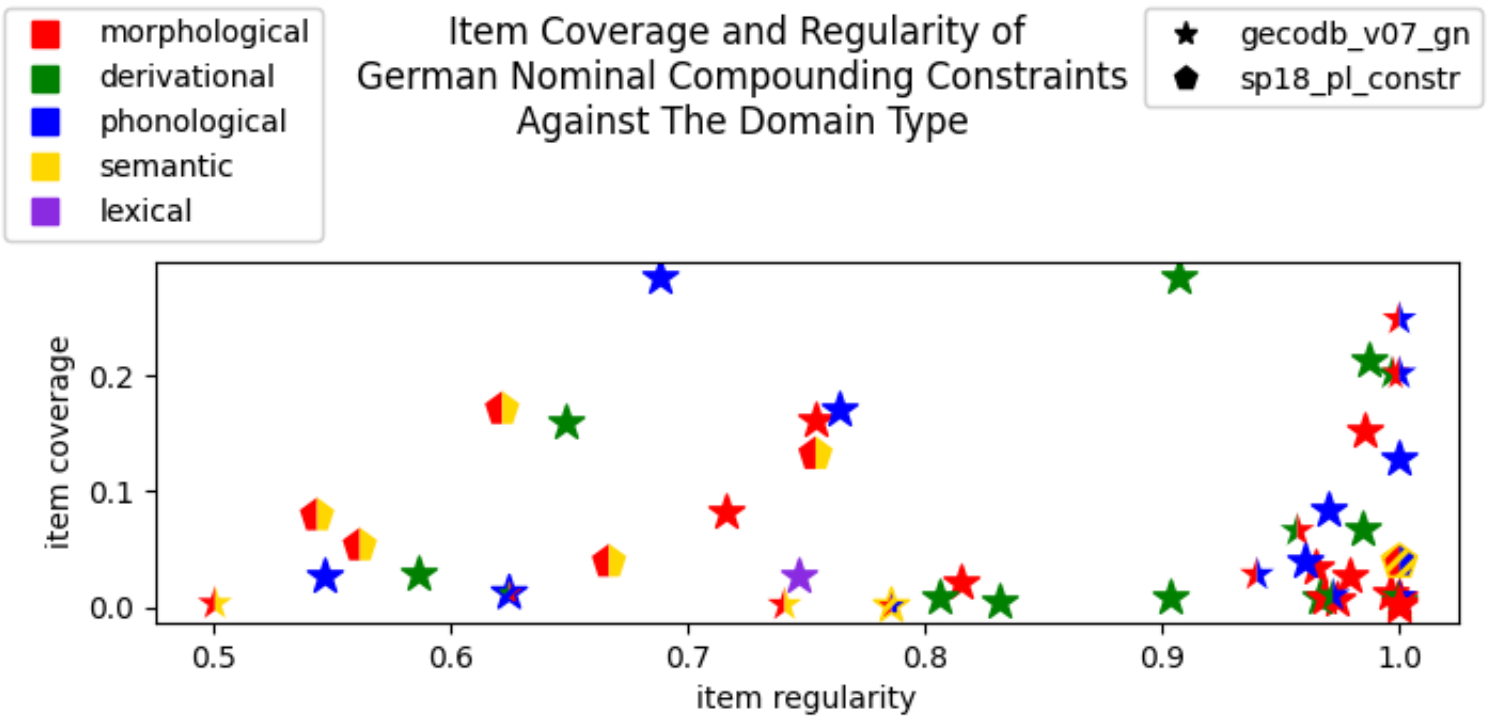
| | item | type |
|---|---|---|
| r | -0.135 | -0.319 |

4. We find that including semantic conditions to constraint definitions statistically decreases the item regularity of the constraints (coeff $-0.2$$-0.2$ at intercept $0.83$$0.83$, p-value $0.001$$0.001$), while no such effect is observed for the other domains. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding. Note that almost all non-confirmed constraints (item regularity $<0.5$$<0.5$) are semantic as well which makes the real picture even worse for semantics.

At this step, we will be treating the domain type of constraints as one-hot vectors. For example, a morphological+phonological constraint will be encoded as $[1, 0, 1, 0, 0]$$[1, 0, 1, 0, 0]$, which is decoded as `[+morphological][+derivational][+phonological][+semantic][+lexical]`. These vectors will then be passed to a linear regression model with an Ordinary Least Squares (OLS) loss function.

Let us consider the scatter plot of constraint item regularity against the domain type. We can immediately see that morphological, derivational, and phonological constraints are scattered all over the lot but are still concentrated towards the right edge of the axis. Contrary to that, semantic (including mixed) constraints only populate the range of $[0.5; 0.8]$$[0.5; 0.8]$; the only constraint with a high regularity is morphological+phonological+semantic. The plot suggests that the constraints that include a semantic condition are less regular.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.



To confirm the intuition, we run a linear regression over the item regularity of the constraints vs. their domain type, based on the one-hot vectors we produced above. To concentrate on individual contribution of each domain, we do not include their interaction terms.

The linear regression model confirms our intuition. The model overall model is significant and explains more variation of item regularity than chance (p-value $0.026$$0.026$). In the model, semantics is the only domain that contributes to item regularity significantly (coeff $-0.2$$-0.2$ at intercept $0.83$$0.83$, p-value $0.001$$0.001$). That is, the item regularity of an arbitrary constraint decreases by $0.2$$0.2$ when a semantic condition included, assuming that the other domains of this constraint remain intact. This effect supports our claim that semantics can only serve as a soft clue in German nominal compounding.

```
Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                reg_item   R-squared:                       0.254
Model:                            OLS   Adj. R-squared:                  0.165
Method:                 Least Squares   F-statistic:                     2.855
Date:                Mon, 07 Sep 2026   Prob (F-statistic):             0.0263
Time:                        16:42:47   Log-Likelihood:                 27.070
No. Observations:                  48   AIC:                            -42.14
Df Residuals:                      42   BIC:                            -30.91
Df Model:                           5
Covariance Type:            nonrobust
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                        0.8302      0.065     12.782      0.000       0.699       0.961
morphological[T.True]            0.0822      0.062      1.332      0.190      -0.042       0.207
derivational[T.True]             0.0302      0.061      0.493      0.624      -0.094       0.154
phonological[T.True]             0.0393      0.051      0.767      0.447      -0.064       0.143
semantic[T.True]                -0.2026      0.059     -3.432      0.001      -0.322      -0.083
lexical[T.True]                 -0.0832      0.161     -0.517      0.608      -0.408       0.241
==============================================================================
Omnibus:                        4.690   Durbin-Watson:                   1.563
Prob(Omnibus):                  0.096   Jarque-Bera (JB):                3.947
Skew:                          -0.698   Prob(JB):                        0.139
Kurtosis:                       3.152   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

The domain type of constraints has no effect on their item coverage (p-value of the overall model $0.889$$0.889$, none of the individual domains are significant).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                cvg_item   R-squared:                       0.038
Model:                             OLS   Adj. R-squared:                 -0.076
Method:                  Least Squares   F-statistic:                    0.3350
Date:                 Mon, 07 Sep 2026   Prob (F-statistic):              0.889
Time:                         16:42:47   Log-Likelihood:                 53.072
No. Observations:                   48   AIC:                            -94.14
Df Residuals:                       42   BIC:                            -82.92
Df Model:                            5
Covariance Type:             nonrobust
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                        0.0923      0.038      2.443      0.019       0.016       0.169
morphological[T.True]           -0.0198      0.036     -0.550      0.585      -0.092       0.053
derivational[T.True]            -0.0225      0.036     -0.631      0.531      -0.095       0.049
phonological[T.True]             0.0105      0.030      0.350      0.728      -0.050       0.071
semantic[T.True]                -0.0188      0.034     -0.548      0.586      -0.088       0.050
lexical[T.True]                 -0.0663      0.094     -0.708      0.483      -0.255       0.123
==============================================================================
Omnibus:                        9.358   Durbin-Watson:                   1.165
Prob(Omnibus):                  0.009   Jarque-Bera (JB):                9.795
Skew:                           1.106   Prob(JB):                      0.00746
Kurtosis:                       3.022   Cond. No.                         10.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

## 4. Constraint Support, Productivity & Regularity

### Description

In this section, we introduce the new (irregularity) scale and apply it to our results.

1. We analyze the productivity, regularity, and support level of the constraints. Most of the constraints are productive and irregular to a different extent.
2. We that manually analyze partially or not supported constraints and find that no constraints require changes in their conditions.

**NB!** In this chapter, we take a look at the entire set of constraints, whether they are confirmed or not.

### Results

- The new (ir)regularity scale did not affect the support level of previously verified constraints much: only four constraints from Shmalts 2026 downgraded to partially support at the new scale. The new *GeCoDB* versions produce an even better result on the old constraints due to cleaner data. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*. On top of that, the changed checkers for the constraints triggered by morphological and phonological complexity of the N1 led to these two constraints being unsupported and partially supported, respectively.
- Ten constraints out of 48 turn out to be unproductive by the Tolerance Principle. Regular constraints by the ITP constitute half of the constraint set: 24 out of 48.
- We find that no constraints require a change of definition. Only regularity levels should be adjusted for them.

Yang 2016 formulates the Tolerance Principle that describes the maximum number of exceptions that a productive rule (in a broader meaning of a linguistic generalization) can tolerate:

$$ {\theta}_N < \frac{N}{ln N} $$$$ {\theta}_N < \frac{N}{ln N} $$

where $N$$N$ is the number of items that a rule is potentially applicable to, and ${\theta}_N$${\theta}_N$ is the maximum tolerable number of exceptions. Each rule, be it productive or not, is defined over a subset of items with certain linguistic properties, identically to our constraints. Thus, the whole relevant vocabulary is partitioned into subsets with certain linguistic properties, and each subset has its own size $N$$N$ and its own (un)productive rule defined over it. The crucial difference between productive and unproductive rules is then, that productive rules are applied automatically to any novel input that satisfies certain linguistic conditions, while unproductive rules are "frozen" and are bound to a certain set of items. Yang 2016 provides numerous evidence from psycholinguistic experiments of language acquisition in support of this distinction. Thus, children cross-linguistically readily overgeneralize productive but virtually never unproductive rules.

Yang 2016 claims that linguistic phenomena are governed by sets of productive and unproductive rules that are applied following thr Elsewhere Condition. Thus, the item input is first tested against the target linguistic properties of the first rule in the set. If the conditions are satisfied, the rule is applied and the derivation stops. If not, the conditions of the second rules applied and so on, until a matching rule is found or until the derivation reaches the last, least specific, default rule. On the example of German plural compounding, Yang 2016 demonstrates that German nouns can be partitioned such that each portion has a productive plural formation rule as tested against the Tolerance Principle. Thus, feminine nouns productively build a plural with *-en*, (non-feminine) nouns ending in schwa as well, then (non-feminine) nouns with condition X build a plural with y, etc. The distinction between productive and unproductive rules is important insofar that the unproductive rules come first in the Elsewhere Condition. How the unproductive and productive rules are ordered between themselves is an open question. Yang 2016 support this hypothesis with evidence from psycholinguistic experiments, too.

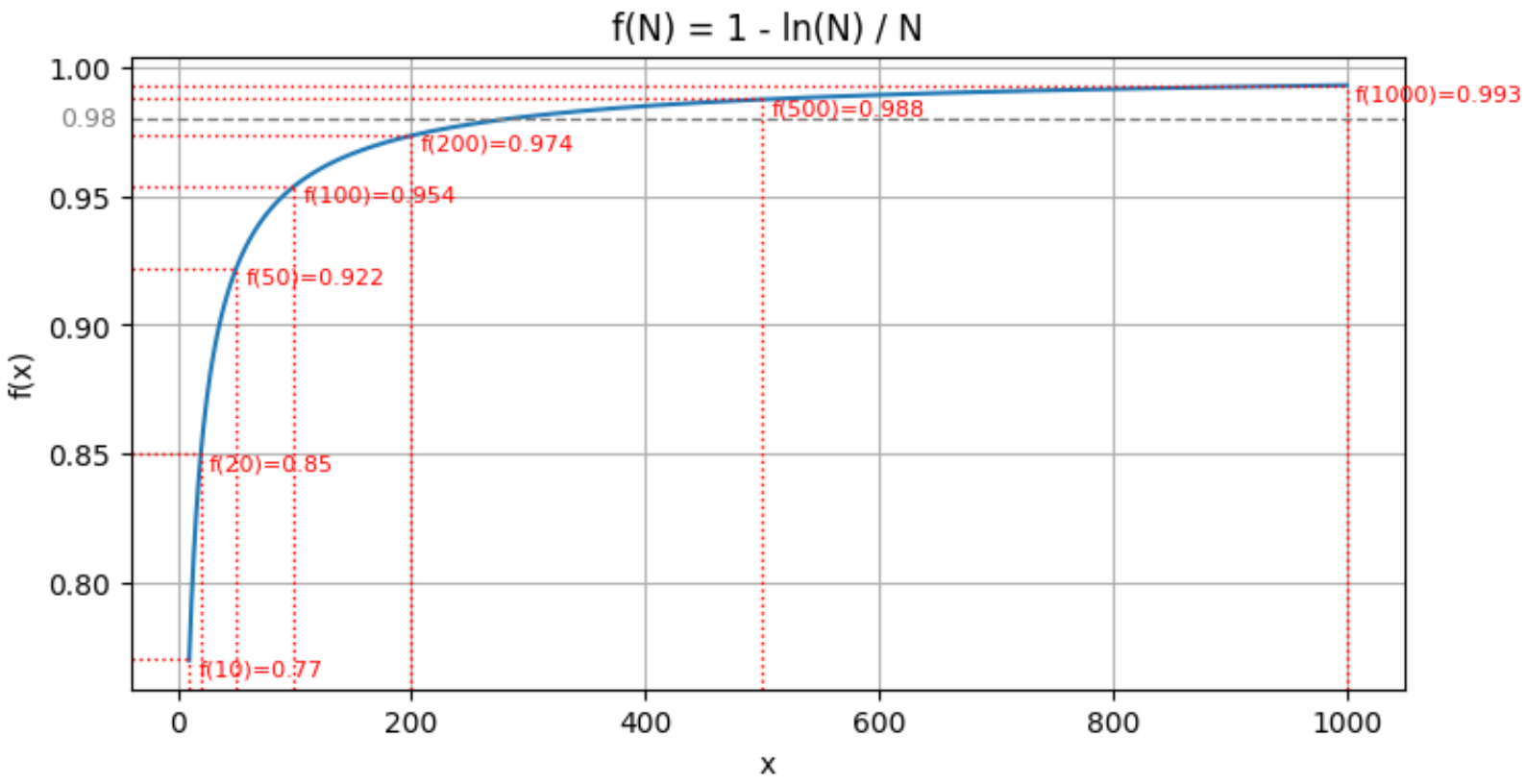
While we will try to model German nominal compounding with Yang's method, it will not be part of this study. This introduction into the Tolerance Principle was given to justify its validity or at least utility. We employ this numeric expression to derive a flexible constraint regularity threshold that accounts for its absolute item coverage. We find that a constant value may become unreasonable when the absolute item coverage is either too low or too high. In the original study, we choose $0.98$$0.98$ as the threshold for regularity. However, if we fit this threshold onto a rule that is potentially applicable to 3000 items, the rule will still be considered regular despite having 60 exceptions which is a lot for a generalization that should be applicable to virtually any input. On the other hand, there are a lot of constraints with an absolute coverage of 20-30 constraints. At this magnitude, $0.98$$0.98$ doesn't even make sense. We find that the multiplicative inverse of the Tolerance Principle threshold is a gut estimate of the maximum acceptable share of violations for a regular constraint. We propose the Regularity Tolerance Principle: a constraint (rule in Yang's terminology) is regular if for its item regularity $\text{reg}$$\text{reg}$ it holds:

$$ \text{reg} \geq 1 - \frac{1}{{\theta}_N} = 1 - \frac{ln N}{N} $$$$ \text{reg} \geq 1 - \frac{1}{{\theta}_N} = 1 - \frac{ln N}{N} $$

On another note, it is not quite clear how to distinguish actual grammatical constraints from mere statistical patterns. Maybe the Tolerance Principle may offer its service here.

Below is the visualization of the Regular Tolerance Principle and thresholds for different absolute item coverage values.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.




Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | TP | ITP |
|---|---|---|
| **10** | 4 | 0.770 |
| **20** | 6 | 0.850 |
| **50** | 12 | 0.922 |
| **100** | 21 | 0.954 |
| **200** | 37 | 0.974 |
| **500** | 80 | 0.988 |
| **1000** | 144 | 0.993 |
| **5000** | 587 | 0.998 |

Based on the results of chapter 3 and the Regularity Tolerance Principle, we propose a new (ir)regularity scale. The scale has two layers. The first layer distinguishes regular from irregular constraints, governed by the Regularity Tolerance Principle (RTP). If the constraint is irregular, we define it (ir)regularity level by the thresholds of $0.7$$0.7$ and $0.9$$0.9$. The layered scale is required to account for small-scale constraints, for which the regularity threshold by the RTP may be below $0.9$$0.9$. The procedure of determining the support level of constraints remains that same as in the original study. Reminder: we declare any constraints with an item regularity of $<0.5$$<0.5$ not supported.

1. The new (ir)regularity scale did not affect the support level of previously verified constraints much. Only four constraints from Shmalts 2026 are now partially supported. The newer versions of *GeCoDB* are even better with three and two partially supported constraints. Two out of five new constrains verified by *GeCoDB_v07_GN* are not supported. Neither are two out of twelve pluralic constraints from *SP18_Pl_Constr*. Note that some of the constraints had zero coverage in all their datasets.

Ten constraints out of 49 turn out to be unproductive by the Tolerance Principle. To estimate the number of exceptions per item, we added up the violation rate of each covered item (this is the same violation rate we used to calculate item regularity). For example, if a dummy constraint covers items A, B, and C, and they build compounds that satisfy the constraint in $0.9$$0.9$, $0.35$$0.35$, and $0.77$$0.77$ cases, respectively, the item regularity of this constraint is $\frac{0.9 + 0.35 + 0.77}{3} = 0.673$$\frac{0.9 + 0.35 + 0.77}{3} = 0.673$, and the number of violations is $(1 - 0.9) + (1 - 0.35) + (1 - 0.77) = 0.98$$(1 - 0.9) + (1 - 0.35) + (1 - 0.77) = 0.98$.

Regular constraints by the RTP constitute slightly less than a half of the constraint set: 24 out of 49.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | gecodb_v06 | gecodb_v07 | gecodb_v07_gn | sp18_pl_constr |
|---|---|---|---|---|
| **supported+** | 12 | 12 | 13 | 4 |
| **supported** | 21 | 22 | 25 | 5 |
| **partially supported** | 5 | 4 | 4 | 0 |
| **not supported** | 1 | 1 | 2 | 2 |

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| | | N |
|---|---|---|
| is_productive | is_regular | |
| **False** | **False** | 10 |
| **True** | **False** | 14 |
| | **True** | 24 |

```
# unproductive constraints
```

| | cvg_item_abs | n_violations_allowed | n_violations | reg_item | actual_reg_level |
|---|---|---|---|---|---|
| constr_id | | | | | |
| **p2l:decl_cl:pl:#e-0ldef-0** | 537.0 | 85.428 | 131.933 | 0.754 | moderate regularity |
| **p2l:decl_cl:pl:#er-0/erlanim-er** | 10.0 | 4.343 | 4.989 | 0.501 | low regularity |
| **p2l:decl_cl:pl:#e_uml-0ldef-0** | 274.0 | 48.814 | 77.935 | 0.716 | moderate regularity |
| **p2l:drv:deverb_schwa$-0ldef-0** | 94.0 | 20.690 | 38.801 | 0.587 | low regularity |
| **p2l:drv:prx_deverb-s** | 527.0 | 84.089 | 184.806 | 0.649 | low regularity |
| **p2l:phon_fin:F_stem_#t$-s** | 41.0 | 11.041 | 15.386 | 0.625 | low regularity |
| **p2l:phon_fin:schwa$-enldef-en** | 569.0 | 89.693 | 134.177 | 0.764 | moderate regularity |
| **l2p:sltend_cmpx_phon** | 1021.0 | 147.362 | 319.000 | 0.688 | low regularity |
| **l2p:sltend_!sonority** | 95.0 | 20.861 | 43.000 | 0.547 | low regularity |
| **l2p:slsmpx_high_freq** | 95.0 | 20.861 | 24.000 | 0.747 | moderate regularity |

2. We now qualitatively inspect partially or not supported constraints. We find that no constraints require changes in their conditions. Only regularity levels should be adjusted for them.

### GeCoDB_v06

Three* partially supported original constraints, one did not reach assumed regularity, and two -- a moderate regularity level.

On top of that, we moved three -s-related tendencies (morphological complexity, phonological complexity, sonority) towards the L2P constraints, and changed the checkers for them correspondingly.This led to one not supported and one partially supported novel constraints.

* `l2p:n` the last is ignored, see below.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| constr_id | appl_item | n_appl_items | quantifier | cvg_item_abs | cvg_item | cvg_type | reg_item | reg_ty |
|---|---|---|---|---|---|---|---|---|
| **p2l:drv:sfx:sfx-sldef-s** | n1 | 3614.0 | regularly | 777.0 | 0.215 | 0.158 | 0.980 | 0. |
| **p2l:drv:prx_deverb-s** | n1 | 3614.0 | tendency | 575.0 | 0.159 | 0.122 | 0.641 | 0. |
| **l2p:sltend_cmpx_phon** | n1+linker | 4870.0 | tendency | 1314.0 | 0.270 | 0.256 | 0.607 | 0. |
| **l2p:sltend_!sonority** | n1+linker | 4870.0 | tend | 242.0 | 0.050 | 0.032 | 0.413 | 0. |

*First constituents that are constituted by derived nouns with suffixes -(ig)keit, -heit, -schaft, -ung, -sal, -ing, -ling, -tum, -um, -ion, also -ität and its allomorphs attach -s- regularly.*

Expected regularity level: **regular**

Actual regularity level: **high regularity**

Verdict: **partially supported**

This constraint is a good example of when the threshold of $0.98$$0.98$ becomes too permissive for a large-scale constraint. While the constrait reaches an item regularity of exactly $0.98$$0.98$, there are 121 N1s that occasionally violate the constraint. Five of those always build violent compounds (at least 10 each), and 21 N1s build ten or more exceptions including 'Generation' ('Generation-en-konflikt'), 'Spezialität' ('Spezialität-en-handel') etc.

With the absolute item coverage of $777$$777$ N1s, the constraint should reach an item regularity of $0.991$$0.991$ to be regular (actual: $0.98$$0.98$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | p2l:drv:sfx:sfx-s\|def-s_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **generation_+en_phänomen** | generation | 123 | 89 | 0.724 |
| **minderheit_+en_beteiligung** | minderheit | 86 | 50 | 0.581 |
| **wirtschaft_plan** | wirtschaft | 683 | 49 | 0.072 |
| **spezialität_+en_laden** | spezialität | 33 | 33 | 1.000 |
| **süßigkeit_+en_fabrik** | süßigkeit | 26 | 26 | 1.000 |
| **...** | ... | ... | ... | ... |
| **genehmigung_verfahren** | genehmigung | 78 | 1 | 0.013 |
| **kompression_strumpf** | kompression | 20 | 1 | 0.050 |
| **kleidung_stück** | kleidung | 58 | 1 | 0.017 |
| **rehabilitation_sport** | rehabilitation | 54 | 1 | 0.019 |
| **eigentum_schutz** | eigentum | 92 | 1 | 0.011 |

121 rows × 4 columns

*First constituents that are constituted by prefixed deverbal nouns exhibit a tendency to attach -s-.*

Expected regularity level: **moderate regularity**

Actual regularity level: **low regularity**

Verdict: **partially supported**

Numerous exceptions overall, 178 N1s that always build violent compounds.

To reach the moderate regularity level, the constraint should reach an item regularity of $0.7$$0.7$ (actual: $0.641$$0.641$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | p2l:drv:prx_deverb-s_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **ersatz_kanal** | ersatz | 423 | 423 | 1.000 |
| **zusatz_abkommen** | zusatz | 360 | 360 | 1.000 |
| **einsatz_strecke** | einsatz | 291 | 291 | 1.000 |
| **anlage_land** | anlage | 287 | 287 | 1.000 |
| **abschluss_tournee** | abschluss | 259 | 259 | 1.000 |
| **...** | ... | ... | ... | ... |
| **auftrieb_hilfe** | auftrieb | 11 | 1 | 0.091 |
| **vorbehalt_los** | vorbehalt | 12 | 1 | 0.083 |
| **befall_stelle** | befall | 13 | 1 | 0.077 |
| **erwerb_los** | erwerb | 91 | 1 | 0.011 |
| **erfahrung_austausch** | erfahrung | 74 | 1 | 0.014 |

317 rows × 4 columns

*Many simplex nouns that constitute first constituents that attach -s- have a high token frequency.*

Expected regularity level: **moderate regularity**

Actual regularity level: **low regularity**

Verdict: **partially supported**

Eighty-five lower-frequency N1s that attach *-s-*.

To reach the moderate regularity level, the constraint should reach an item regularity of $0.7$$0.7$ (actual: $0.649$$0.649$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | l2p:s\|smpx_high_freq_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **fasching_+s_turnier** | fasching+s | 129 | 129 | 1.0 |
| **heirat_+s_frage** | heirat+s | 75 | 75 | 1.0 |
| **adel_+s_gesellschaft** | adel+s | 68 | 68 | 1.0 |
| **hering_+s_art** | hering+s | 46 | 46 | 1.0 |
| **esel_+s_treiber** | esel+s | 45 | 45 | 1.0 |
| **...** | ... | ... | ... | ... |
| **notar_+s_kammer** | notar+s | 1 | 1 | 1.0 |
| **brei_+s_gau** | brei+s | 1 | 1 | 1.0 |
| **sumpf_+s_egge** | sumpf+s | 1 | 1 | 1.0 |
| **speck_+s_winkel** | speck+s | 1 | 1 | 1.0 |
| **zauber_+s_pass** | zauber+s | 1 | 1 | 1.0 |

85 rows × 4 columns

*All nouns that constitute first constituents that attach the -n- allomorph of the -(e)n- linker end in schwa or in [ɐ] (unstressed -er).*

Seven out of eight violent N1s are misanalyses and are removed in further *GeCoDB* versions:

- misanalyzed conversions
  - zwiebel_+n_schneide: beim Zwiebelnschneiden
  - windel_+n_wechsel: beim Windelnwechseln
  - vokabel_+n_pauke: beim Vokabelpauken
  - waffel_+n_backe: beim Waffelnbacken
- proper names
  - mittel_+n_dorf, mittel_+n_kirche
  - nachbar_+n_land

Therefore, this result is ignored and declared **supported**.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | l2p:n_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **mittel_+n_kirche** | mittel+en | 2 | 2 | 1.0 |
| **nachbar_+n_land** | nachbar+en | 1 | 1 | 1.0 |
| **zwiebel_+n_schneide** | zwiebel+en | 1 | 1 | 1.0 |
| **windel_+n_wechsel** | windel+en | 1 | 1 | 1.0 |
| **vokabel_+n_pauke** | vokabel+en | 1 | 1 | 1.0 |
| **waffel_+n_backe** | waffel+en | 1 | 1 | 1.0 |
| **semmel_+n_knödel** | semmel+en | 1 | 1 | 1.0 |

The changed checkers for the tendency constraints moved towards the L2P constraints.

*The probability of -s- exhibits a tendency to grow with the phonological complexity of the first constituent. By phonologically complex words are understood words with polysyllable non-trochaic form, words with unstressed prefixes, words with stressed or semi-stressed suffixes etc.*

Expected regularity level: **moderate regularity**

Actual regularity level: **low regularity**

Verdict: **partially supported**

An huge portion of violations: trochaic N1 that attach -s-.

To reach the moderate regularity level, the constraint should reach an item regularity of $0.7$$0.7$ (actual: $0.607$$0.607$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | l2p:s\|tend_cmpx_phon_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **arbeit_+s_trupp** | arbeit+s | 706 | 706 | 1.0 |
| **wirtschaft_+s_berechnung** | wirtschaft+s | 634 | 634 | 1.0 |
| **leben_+s_kerze** | leben+s | 627 | 627 | 1.0 |
| **liebling_+s_stand** | liebling+s | 592 | 592 | 1.0 |
| **bildung_+s_innovation** | bildung+s | 532 | 532 | 1.0 |
| **...** | ... | ... | ... | ... |
| **steg_+s_kopf** | steg+s | 1 | 1 | 1.0 |
| **flug_+s_pass** | flug+s | 1 | 1 | 1.0 |
| **ziel_+s_dorf** | ziel+s | 1 | 1 | 1.0 |
| **speck_+s_winkel** | speck+s | 1 | 1 | 1.0 |
| **sumpf_+s_egge** | sumpf+s | 1 | 1 | 1.0 |

516 rows × 4 columns

*The probability of -s- in compounds with simplex first constituents tends to decrease with the increasing sonority of the final segment of the first constituent. -s- is thus more frequent after plosives, infrequent after nasals and liquids, and it never occurs after a full vowel.*

Expected regularity level: **moderate regularity**

Actual regularity level: **below scale**

Verdict: **not supported**

An enormous portion violations: liquid-/nasal-final N1s that attach -s-.

To reach the moderate regularity level, the constraint should reach an item regularity of $0.7$$0.7$ (actual: $0.388$$0.388$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | l2p:s\|tend_!sonority_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **frieden_+s_hetzer** | frieden+s | 232 | 232 | 1.0 |
| **himmel_+s_fee** | himmel+s | 223 | 223 | 1.0 |
| **karneval_+s_schmuck** | karneval+s | 194 | 194 | 1.0 |
| **schaden_+s_minderung** | schaden+s | 166 | 166 | 1.0 |
| **teufel_+s_zimmer** | teufel+s | 130 | 130 | 1.0 |
| **...** | ... | ... | ... | ... |
| **aal_+s_meer** | aal+s | 1 | 1 | 1.0 |
| **nickel_+s_dorf** | nickel+s | 1 | 1 | 1.0 |
| **monster_+s_pass** | monster+s | 1 | 1 | 1.0 |
| **garn_+s_dorf** | garn+s | 1 | 1 | 1.0 |
| **regen_+s_dorf** | regen+s | 1 | 1 | 1.0 |

142 rows × 4 columns

## GeCoDB_v07

The same three partially supported original constraints as with *GeCoDB_v06*; the same two changed tendency constraints; see details here.

As discussed in the case of `l2p:n`, all but one exceptions were mysanalises. After removing those from the data, the constraint is regular and supported.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| constr_id | appl_item | n_appl_items | quantifier | cvg_item_abs | cvg_item | cvg_type | reg_item | reg_ty |
|---|---|---|---|---|---|---|---|---|
| **p2l:drv:sfx:sfx-sldef-s** | n1 | 3593.0 | regularly | 777.0 | 0.216 | 0.160 | 0.980 | 0. |
| **p2l:drv:prx_deverb-s** | n1 | 3593.0 | tendency | 574.0 | 0.160 | 0.124 | 0.643 | 0. |
| **l2p:sltend_cmpx_phon** | n1+linker | 4741.0 | tendency | 1252.0 | 0.264 | 0.259 | 0.637 | 0. |

## GeCoDB_v07_GN

Two out of three* partially supported constraints from *GeCoDB_v06* and *GeCoDB_v07*; the same two changed tendency constraints (even though now the sonority is partially supported); see details there. Novel constraints: two semantical constraints that appear to be below the scale even despite having viable absolute coverages of $31$$31$ and $28$$28$, respectively.

* `l2p:s|smpx_high_freq` is reasonably supported in *GeCoDB_v07_GN* because GermaNet has less low-frequent unidiomatic compounds.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| constr_id | appl_item | n_appl_items | quantifier | cvg_item_abs | cvg_item | cvg_type | reg_item | re |
|---|---|---|---|---|---|---|---|---|
| p2l:decl_cl:pl:#e-0lmsyl_anim-e | n1 | 3319.0 | irregularly | 31.0 | 0.009 | 0.009 | 0.088 | |
| p2l:drv:sfx:sfx-sldef-s | n1 | 3319.0 | regularly | 710.0 | 0.214 | 0.154 | 0.987 | |
| p2l:drv:prx_deverb-s | n1 | 3319.0 | tendency | 527.0 | 0.159 | 0.123 | 0.649 | |
| p2l:sem:2const_anim/pers-s | n1+n2 | 48660.0 | tend | 28.0 | 0.001 | 0.001 | 0.482 | |

*First constituents that are constituted by monosyllabic animal designations that build the plural form with -e attach -e- irregularly.*

Expected regularity level: **low regularity**

Actual regularity level: **below scale**

Verdict: **not supported**

Out of $31$$31$ covered N1s, $28$$28$ produce only violent compounds, and two more -- a few more violent compounds, which leads to a drastically low item regularity of $0.088$$0.088$. The constraint is destroyed by the data.

The violent N1s includes animals, fish, and birds. Possible distinctions between mammals and non-mammals or that between domesticated and non-domesticated animals are not confirmed: 'Schaf', 'Stier', 'Ross' always produce violent compounds. 'Hund' and 'Schwein' are the only N1s that satisfy the constraint more or less regularly.

To reach the low regularity level, the constraint should reach an item regularity of $0.5$$0.5$ (actual: $0.088$$0.088$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | p2l:decl_cl:pl:#e-0\|msyl_anim-e_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **fisch_schwarm** | fisch | 83 | 83 | 1.000 |
| **krebs_bekämpfung** | krebs | 27 | 27 | 1.000 |
| **schaf_züchter** | schaf | 22 | 22 | 1.000 |
| **hirsch_geweih** | hirsch | 16 | 16 | 1.000 |
| **wal_öl** | wal | 12 | 12 | 1.000 |
| **schwein_+s_galopp** | schwein | 43 | 10 | 0.233 |
| **schmetterling_+s_art** | schmetterling | 9 | 9 | 1.000 |
| **lachs_häppchen** | lachs | 7 | 7 | 1.000 |
| **stier_blut** | stier | 7 | 7 | 1.000 |
| **aal_bestand** | aal | 6 | 6 | 1.000 |
| **ross_haar** | ross | 5 | 5 | 1.000 |
| **hering_+s_möwe** | hering | 5 | 5 | 1.000 |
| **kuckuck_+s_nest** | kuckuck | 5 | 5 | 1.000 |
| **mammut_baum** | mammut | 5 | 5 | 1.000 |
| **reh_kitz** | reh | 5 | 5 | 1.000 |
| **kamel_haar** | kamel | 4 | 4 | 1.000 |
| **elch_fleisch** | elch | 4 | 4 | 1.000 |
| **specht_vogel** | specht | 4 | 4 | 1.000 |
| **hai_attacke** | hai | 4 | 4 | 1.000 |
| **thunfisch_dose** | thunfisch | 3 | 3 | 1.000 |
| **habicht_+s_kraut** | habicht | 3 | 3 | 1.000 |
| **dachs_jagd** | dachs | 3 | 3 | 1.000 |
| **hund_+s_kamille** | hund | 69 | 3 | 0.043 |
| **nerz_fell** | nerz | 3 | 3 | 1.000 |
| **schädling_+s_befall** | schädling | 2 | 2 | 1.000 |
| **krokodil_+s_träne** | krokodil | 2 | 2 | 1.000 |
| **sperling_+s_vogel** | sperling | 2 | 2 | 1.000 |
| **luchs_auge** | luchs | 1 | 1 | 1.000 |
| **kranich_vogel** | kranich | 1 | 1 | 1.000 |
| **hengst_fohlen** | hengst | 1 | 1 | 1.000 |

*Compounds whose first constituent designates a person and whose second constituent is one of 'Mann', 'Frau', 'Leute', 'Tochter', 'Gattin', or 'Witwe' tend to insert -s-.*

Expected regularity level: **low regularity**

Actual regularity level: **below scale**

Verdict: **not supported**

Out of $28$$28$ covered N1+N2 combinations, exactly half violates the constraint by building a single possible violent variant, and N1+N2 'Bürger' + 'Tochter' does so by building a violent compound 'Bürgertochter' alongside with 'Bürgerstochter'. Interestingly, all violations seem to have been defeated by another constraints: mostly the *-er/-el* or schwa constraints.

To reach the low regularity level, the constraint should reach an item regularity of $0.5$$0.5$ (actual: $0.482$$0.482$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | p2l:sem:2const_anim/pers-s_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **kind_+er_frau** | kind+frau | 1 | 1 | 1.0 |
| **bürger_tochter** | bürger+tochter | 2 | 1 | 0.5 |
| **kontakt_mann** | kontakt+mann | 1 | 1 | 1.0 |
| **spieler_frau** | spieler+frau | 1 | 1 | 1.0 |
| **arzt_tochter** | arzt+tochter | 1 | 1 | 1.0 |
| **arbeiter_frau** | arbeiter+frau | 1 | 1 | 1.0 |
| **kaiser_tochter** | kaiser+tochter | 1 | 1 | 1.0 |
| **schütze_+n_frau** | schütze+frau | 1 | 1 | 1.0 |
| **unternehmer_frau** | unternehmer+frau | 1 | 1 | 1.0 |
| **pate_+n_tochter** | pate+tochter | 1 | 1 | 1.0 |
| **krieger_witwe** | krieger+witwe | 1 | 1 | 1.0 |
| **kanzler_gattin** | kanzler+gattin | 1 | 1 | 1.0 |
| **pastor_+en_tochter** | pastor+tochter | 1 | 1 | 1.0 |
| **enkel_tochter** | enkel+tochter | 1 | 1 | 1.0 |
| **patrizier_tochter** | patrizier+tochter | 1 | 1 | 1.0 |

## SP18_Pl_Constr

Two pluralic constraints below scale. Please note that a few pluralic constraints are excluded from here because they have an absolute item coverage of $\leq 3$$\leq 3$. From the two remaining unsupported constraints, one has a an absolute item coverage of $4$$4$ which may also require revision on another dataset.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| constr_id | appl_item | n_appl_items | quantifier | cvg_item_abs | cvg_item | cvg_ty |
|---|---|---|---|---|---|---|
| **p2l:phon_fin:weak_F_cons$-0/s/enl!pl_interpr-0/s** | n1+sem | 76.0 | mostly | 4.0 | 0.053 | 0.0 |

*First constituents that are constituted by consonant-final weak feminine nouns mostly adopt a zero linker or -s- when they have a singular meaning.*

Expected regularity level: **low regularity**

Actual regularity level: **below scale**

Verdict: **not supported**

Out of mere $4$$4$ covered N1s with plural semantics, three build a non-negligible number of exceptions, having a violation rate of $\geq 0.5$$\geq 0.5$ each.0

To reach the low regularity level, the constraint should reach an item regularity of $0.5$$0.5$ (actual: $0.41$$0.41$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | p2l:phon_fin:weak_F_cons$-0/s/en\|!pl_interpr-0/s_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| person_+en_datum | person+sgsem | 46 | 44 | 0.957 |
| schwester_+n_tag | schwester+sgsem | 21 | 11 | 0.524 |
| nation_+en_wertung | nation+sgsem | 7 | 6 | 0.857 |

*Linkers that are identical to the plural ending of the first constituent (except for -s- and zero linker) in a given compound tend to be associated with a plural meaning of this first constituent within the compound.*

Expected regularity level: **moderate regularity**

Actual regularity level: **below scale**

Verdict: **not supported**

Very productive violation. All $40$$40$ covered N1s with a plural linker violate the constraint, from which $39$$39$ have a violation rate of $\geq 0.5$$\geq 0.5$, $33$$33$ of $\geq 0.75$$\geq 0.75$, and $12$$12$ always violent.

To reach the moderate regularity level, the constraint should reach an item regularity of $0.7$$0.7$ (actual: $0.146$$0.146$).

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

| comp_gecodb | l2p:pl_marker\|def-pl_item | Nbwi_total | Nbwi_exc | %_exc |
|---|---|---|---|---|
| **sonne_+n_liege** | sonne+en | 53 | 53 | 1.000 |
| **kind_+er_lied** | kind+er | 59 | 53 | 0.898 |
| **katze_+n_baby** | katze+en | 50 | 50 | 1.000 |
| **hund_+e_lehrer** | hund+e | 54 | 50 | 0.926 |
| **bad_+=er_mann** | bad+"er | 53 | 47 | 0.887 |
| **ei_+er_uhr** | ei+er | 47 | 46 | 0.979 |
| **person_+en_kontrolle** | person+en | 58 | 44 | 0.759 |
| **weg_+e_beziehung** | weg+e | 50 | 44 | 0.880 |
| **frau_+en_kleid** | frau+en | 60 | 44 | 0.733 |
| **buch_+=er_verleih** | buch+"er | 54 | 43 | 0.796 |
| **gerät_+e_version** | gerät+e | 51 | 42 | 0.824 |
| **gitarre_+n_begleitung** | gitarre+en | 42 | 41 | 0.976 |
| **bild_+er_show** | bild+er | 49 | 40 | 0.816 |
| **ohr_+en_weide** | ohr+en | 37 | 36 | 0.973 |
| **geschenk_+e_rausch** | geschenk+e | 36 | 30 | 0.833 |
| **stadt_+=e_ziel** | stadt+"e | 42 | 27 | 0.643 |
| **bett_+en_hersteller** | bett+en | 34 | 26 | 0.765 |
| **haus_+=er_modell** | haus+"er | 36 | 24 | 0.667 |
| **lied_+er_abend** | lied+er | 32 | 24 | 0.750 |
| **mutter_+=_rente** | mutter+" | 31 | 24 | 0.774 |
| **brett_+er_boden** | brett+er | 22 | 21 | 0.955 |
| **vater_+=_sitte** | vater+" | 20 | 17 | 0.850 |
| **schloss_+=er_direktor** | schloss+"er | 18 | 15 | 0.833 |
| **schwester_+n_buch** | schwester+en | 15 | 11 | 0.733 |
| **hemd_+en_tasche** | hemd+en | 10 | 9 | 0.900 |
| **nation_+en_begriff** | nation+en | 8 | 6 | 0.750 |
| **hand_+=e_desinfektion** | hand+"e | 9 | 6 | 0.667 |
| **kommune_+n_grenze** | kommune+en | 5 | 5 | 1.000 |
| **schwert_+er_klang** | schwert+er | 4 | 4 | 1.000 |
| **apfel_+=_säure** | apfel+" | 3 | 3 | 1.000 |
| **rad_+=er_rolle** | rad+"er | 8 | 3 | 0.375 |
| **loch_+=er_stopfer** | loch+"er | 2 | 2 | 1.000 |
| **ball_+=e_becken** | ball+"e | 3 | 2 | 0.667 |
| **nagel_+=_mal** | nagel+" | 1 | 1 | 1.000 |
| **geräusch_+e_macher** | geräusch+e | 1 | 1 | 1.000 |
| **brief_+e_schreiber** | brief+e | 2 | 1 | 0.500 |
| **vogel_+=_häuschen** | vogel+" | 1 | 1 | 1.000 |
| **wurm_+=er_fresser** | wurm+"er | 1 | 1 | 1.000 |
| **zahn_+=e_putzer** | zahn+"e | 1 | 1 | 1.000 |
| **zitat_+e_sammlung** | zitat+e | 1 | 1 | 1.000 |